

PRE-BOARD EXAMINATION

Level: Bachelor
Program: BCSIT
Semester: III
Course: Database Management System

(Semester-Fall)

Year: 2025
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are expected to answer the questions in their own words as far as practicable. The figures in the margin indicate full marks.

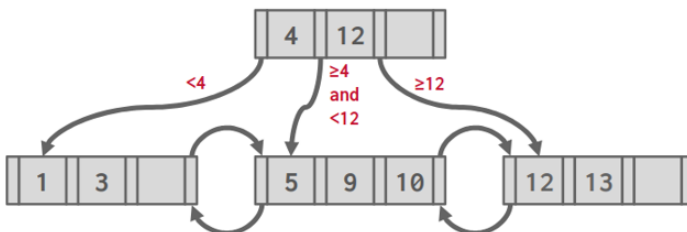
Section “A”**Very Short Answer Questions****Attempt all the questions [10×2=20]**

1. List types of attributes in ER model.
2. Explain left join with Venn Diagram
3. What are aggregate functions in SQL? Explain with an example.
4. List the process involved in query processing.
5. Explain the Projection operation in relational algebra with an example.
6. Write the relational algebra expression for
select name,address from employee where id=5
7. What are pages in database?
8. What is encryption?
9. What is a transaction?
10. List the different types of failures that can occur in a database system.

Section “B”**Descriptive Answer Questions****Attempt five questions [5×10=50]**

11. What are the different types of attributes in ER model? Explain with examples.
12. What are the steps in query processing? Explain each of the steps in detail.
13. Discuss the principles of B+ Tree indexing.

Given the tree below with minimum numbers of key=2 and maximum number of key=3. Show the steps for inserting key 6



14. a. Explain the need for database security and concept of access control.
b. Explain what is view and why views enhance security.
15. What is concurrency control and why do we need it? Explain the states of a transaction with a state diagram.
16. What is a Database Management System? What are the advantages of Database System over file base system?

Section “C”

Long Answer Questions

Attempt any Two questions [2×15=30]

17. Design an ER diagram for a CONFERENCE_REVIEW database that manages research papers submitted to a conference. Authors are uniquely identified by email and have first and last names. Each paper has a unique system-assigned ID, a title, an abstract, and a file name, and may have multiple authors, with one designated as the contact author. Reviewers are uniquely identified by email and have a first name, last name, phone number, affiliation, and topics of interest. Each paper is reviewed by two to four reviewers, and for every assigned paper, a reviewer provides ratings (1–10) on technical merit, readability, originality, and relevance, along with an overall recommendation and two types of comments—one for the review committee and one for the authors.
18. Write SQL query for the following.
 - a. Create a table called department with depart_id as primary key. It should also have the depart_name as attribute with appropriate data type.
 - b. Create a table called employee with emp_id as primary key. It should also contain depart_id, emp_name, join_date and salary with appropriate data types. The depart_id field should be foreign key from department table.
 - c. Insert two departments and four employees in department and employee table respectively.
 - d. Get the count of employee by department.
 - e. Find the employees with second highest salary using the window function.
19. a. Explain how the hash-based indexing works in file organization. What are the advantages and disadvantages of using the hash-based indexing?
b. Explain the ACID property of a transaction. How does database ensure each property?

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PRE-BOARD EXAMINATION

Level:	Bachelor	(Semester-Fall)	Year: 2025
Program:	BCSIT		Full Marks: 100
Semester:	III		Pass Marks: 45
Course:	Internet Technology II (Programming)		Time: 3 hrs.

Candidates are expected to answer the questions in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions [10×2=20]

1. Explain the difference between pre-increment and post-increment operators.
2. What is the purpose of the switch statement in PHP? Write its syntax.
3. Differentiate between indexed array and associative array in PHP.
4. Explain the purpose of htmlspecialchars() function in form handling.
5. How do you send emails using PHP? Mention the function used.
6. Differentiate between sessions and cookies.
7. What is a transaction in database? Why is it used?
8. What is the purpose of \$this keyword in PHP classes?
9. List any four magic methods in PHP.
10. Explain the role of Model in MVC architecture.

Section “B”

Descriptive Answer Questions

Attempt five questions [5×10=50]

11. Discuss the steps involved in setting up a PHP development environment using XAMPP. Explain different types of operators with suitable examples.
12. Define loop control structures. Explain the different types of loops with their syntax and examples.
13. Explain the concept of user-defined functions in PHP. Discuss function declaration, passing arguments, return values, and variable scoping with suitable examples.
14. Discuss transaction management in MySQL using PHP. Write a PHP program demonstrating a banking transaction where money is transferred from one account to another.
15. Explain the MVC architecture and its implementation in a PHP framework like Laravel. Discuss the benefits of using such a framework.

16. Write a complete PHP program to implement session-based authentication. Include:
- Registration page
 - Login page
 - Protected home page (accessible only after login)
 - Logout functionality

Section “C”

Long Answer Questions

Attempt any two questions [2×15=30]

17. a. Define Database Management System? Write a PHP script to perform CRUD (Create, Read, Update, Delete) operations on a MySQL database. Include examples of each operation. [10]
- b. Explain file handling in PHP and discuss different file operations. [5]
18. a. Explain the following built-in functions with examples: [5]
- i. array_pop()
 - ii. array_slice()
 - iii. sort()
 - iv. in_array()
 - v. array_search()
- b. How do you handle form submissions in PHP? Illustrate with an example. [10]
19. Discuss the principles of object-oriented programming in PHP, focusing on abstraction and polymorphism. Write a program to demonstrate these principles.

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PRE-BOARD EXAMINATION

Level:	Bachelor	(Semester-Fall)	Year: 2025
Program:	BCSIT		Full Marks: 100
Semester:	III		Pass Marks: 45
Course:	Object Oriented Analysis and Design		Time: 3 hrs.

Candidates are expected to answer the questions in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions [10×2=20]

1. Define relationship between different class with example.
2. How user interaction analysis will help for requirement analysis.
3. What does responsibility mean in OOAD? Provide suitable example
4. What do source message, processing service, and output message represent in a UML sequence diagram? Explain.
5. What is “Designing for Visibility” in software design?
6. Describe the roles and responsibilities of the designer or architect and the developer in project implementation.
7. How does product manufacturing represent the Factory Design Pattern? provide suitable example.
8. Describe the term Design pattern?
9. Differentiate between design and coding.
10. What is the role of class discussion and presentation for case study-based project?

Section “B”

Descriptive Answer Questions

Attempt any Five questions [5×10=50]

11. What are advantages of object-oriented system over procedural system? Explain association, aggregation and composition relationships with suitable example
12. What is requirement. Explain how the requirement elicitation process happens in OOA.
13. Construct a System Sequence Diagram (SSD) for online examination system with necessary assumption.
14. Explain the purpose and benefits of information Expert, Creator, Controller and Polymorphism design patterns defined by GRASP.

15. How can we create class definition and methods from the domain class diagram and interaction diagram? Explain with appropriate example
16. Explain the SOLID principles. Choose any two principles and explain them by identifying a problem and providing an appropriate solution with a suitable example.

Section “C”

Long Answer Questions

Attempt any Two questions [2×15=30]

17. a. Describe functional and nonfunctional requirements with suitable example
b. Discuss the mechanism of transformation from OOA to OOD? prepare the Activity diagram for restaurant booking system.
18. Consider the scenario of flight booking through an online system at first and then with allowance of certain days of delay for final purchase through second round of payment process completion.
 - a. Identify all the actors, use-cases and relationship.
 - b. Also draw use case diagram.
19. a. For the case study given below identify all classes, their relationships, attributes and methods for each class. Also draw class diagram using standard UML Notation.

XYZ Marina is privately owned corporation that rents boats and provides boat services on a lake. The Corporation needs automated system to track customers, leased slips (Each space for boat in lake) and boats in the slips. The corporation has two types of boats sailboat and power boat. Both types of boats are uniquely identified by their attributes like registration number, manufacture year and boat length. The boat can be leased on daily basis or yearly basis. The system should perform following tasks creating lease, computing lease amount, assigning boats. The system should also have features of search for vacant slips leased to specific customers and generating customized reports. The system should have to implement billing system as well.

 - b. For the above class diagram created apply object-oriented techniques to convert such class diagram into implementation code in any of your favorite object-oriented programming language. Your implementation code should clearly show class definition, Attributes with their proper visibility and method signatures with required parameters

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PRE-BOARD EXAMINATION

Level:	Bachelor	(Semester-Fall)	Year: 2025
Program:	BCSIT		Full Marks: 100
Semester:	III		Pass Marks: 45
Course:	Principles of Management		Time: 3 hrs.

Candidates are expected to answer the questions in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions (10×2=20)

1. What are the roles of manager?
2. What are the different theory of classical management theory?
3. List the approaches of decision making?
4. Describe the contemporary issues in planning?
5. Differentiate between Management and Leader?
6. Concept of Motivation?
7. List the types of controlling?
8. Why IT is important for management?
9. Difference between leader and Power
10. Contemporary issues in planning?

Section “B”

Descriptive Answer Questions

Attempt any Five questions (5×10=60)

11. Define Management. What are the contemporary challenges that Nepalese managers confront in 2025? How to manage it? Illustrate your answer with suitable examples.
12. Compare and contrast the Maslow’s need hierarchy, and Herzberg’s dual factor theory of motivation.
13. What are the approaches in decision making? What are the strategies for improving decision making of Nepalese organizations? Explain.
14. Explain Path-goal Theory?
15. What is emotional intelligence in leadership. What are the components of emotional intelligence? Explain.
16. What is effective control. What are the contemporary issues in controlling resources of Nepalese organizations? Explain with proper illustration.

Section “C”
Long Answer Questions

Attempt any Two questions (2×15=30)

17. Explain emerging trends in Information Management?
18. BrightTech Electronics is a medium-sized company that manufactures smart home appliances such as air purifiers, smart lights, and security cameras. The company was started by Mr. Arjun Mehta, who initially managed all major activities himself. As the business grew, BrightTech expanded its operations to multiple cities and hired more employees. Despite increased sales, the company began facing several problems. Employees were confused about their duties, two departments often performed the same tasks, and delays occurred in production and delivery. The sales team blamed the production department for late deliveries, while production staff complained about unrealistic sales targets. Mr. Mehta realized that the company lacked a proper organizational structure. To solve these issues, Mr. Mehta decided to reorganize the company. He grouped activities into separate departments such as Production, Sales, Finance, and Human Resources. Departmental managers were appointed and given clear authority and responsibility. Reporting relationships were defined, and employees were informed about their specific roles. As a result, coordination improved, duplication of work reduced, and overall efficiency increased.

Questions (Typical Exam Style)

- a. Identify the management function discussed in the above case.
 - b. Explain the importance of organizing highlighted in the case.
 - c. Name and explain any two elements of organizing mentioned in the case.
 - d. How did proper organizing help BrightTech Electronics improve efficiency?
19. ***Case Study of Adani Group***
The Adani Group, one of India’s largest multinational conglomerates, operates in diverse sectors such as energy, ports, logistics, airports, mining, infrastructure, and green energy. Founded by Gautam Adani, the group’s rapid growth is largely attributed to its systematic and long-term planning process.

Vision and Objectives

The Adani Group follows a clear vision of nation-building and sustainable growth. Its planning begins with setting long-term objectives, such as

becoming a global leader in infrastructure development and renewable energy. For example, the group planned to invest heavily in green energy projects to support India's renewable energy goals.

Strategic Planning

At the top level, the management formulates strategic plans after analyzing the business environment, government policies, market demand, and technological changes. The decision to expand from ports and power into airports and renewable energy reflects careful strategic planning and risk assessment.

Operational Planning

Once strategies are finalized, middle and lower-level managers prepare operational plans, including budgets, schedules, and resource allocation. For instance, when developing solar and wind energy projects, detailed plans are made regarding land acquisition, technology, workforce, and timelines.

Flexibility and Contingency Planning

The Adani Group also practices flexible planning. During economic fluctuations and regulatory changes, the group revises its plans and adopts contingency measures to reduce risk and ensure continuity of operations.

Outcome

Due to effective planning, the Adani Group has achieved rapid expansion, diversification, and competitive advantage. Proper planning has helped the organization minimize uncertainty, utilize resources efficiently, and achieve long-term sustainability.

- a. Identify the **type of planning** followed by the Adani Group.
- b. How does planning help the Adani Group in achieving its objectives?
- c. Explain the importance of **flexibility in planning** with reference to the case.

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PRE-BOARD EXAMINATION

Level:	Bachelor	(Semester-Fall)	Year: 2025
Program:	BCSIT		Full Marks: 100
Semester:	III		Pass Marks: 45
Course:	Statistics and Probability		Time: 3 hrs.

Candidates are expected to answer the questions in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions (10×2=20)

1. If the values of the lower and upper quartiles are 50 and 90, respectively, then calculate the inter-quartile range and quartile deviation.
2. If $n = 2000$ and $p = 0.05$ in a Poisson distribution, what is the mean and standard deviation?
3. Find the variance of the binomial distribution when $q = 0.6$ and $n = 20$.
4. Importance of graphical representation of data.
5. What are the steps in hypothesis testing?
6. Short note on: variables and their types.
7. Importance of normal distribution.
8. Types of errors in hypothesis testing.
9. A random sample of 36 employees from a company has an average monthly income of NPR 45,000 with a standard deviation of NPR 6,000. Calculate the standard error of the mean.
10. In a municipality, a survey of 400 households found that 260 households use solar energy for lighting. Calculate the standard error of the proportion.

Section “B”

Descriptive Answer Questions

Attempt any Five questions (5×10=50)

11. In a manufacturing plant, each worker has a 1% chance ($p = 0.01$) of experiencing a minor workplace accident in a year. For a large number of workers, accident occurrences may be approximated by a Poisson distribution. Out of 400 workers, find approximately:
 - a) The number of workers who will have no accidents in a year.
 - b) The number of workers who will have fewer than 2 accidents in a year.

12. According to a recent survey, 40% of employees in a company prefer remote working over office-based work. Suppose this proportion is true for the entire employee population and that preferences follow a binomial distribution. If 15 employees are randomly selected, find the probability that:

- a) Fewer than 3 employees prefer remote working.
- b) At least 1 employees prefer remote working.

13. A company wants to examine whether there is a significant difference in mean monthly sales (in thousand NPR) between trained and untrained employees. The following summary statistics are available:

Group	Sample Size	Mean Sales	Standard Deviation
Trained Employees	8	53.5	2.45
Untrained Employees	8	46.9	2.03

At the 5% level of significance, test whether there is a significant difference in mean sales between the two groups.

14. Let 2, 6, 8, and 12 be the 4 population values. If someone randomly selects 2 samples with replacement, a) obtain possible samples, b) calculate the population mean and standard deviation, and c) show that the sample mean is an unbiased estimate of the population mean.
15. The daily electricity consumption of households in a city is normally distributed with a mean of 30 kWh and a standard deviation of 5 kWh.
- a) Find the probability that a randomly selected household consumes less than 25 kWh per day
 - b) Find the probability that a household consumes between 28 kWh and 35 kWh per day
 - c) What proportion of households consume more than 40 kWh per day?
16. A public health researcher wants to estimate the proportion of households using safe drinking water in a municipality that has a total of 12,000 households. Previous studies suggest that about 60% of households use safe drinking water. The researcher wishes to estimate this proportion with a 95% confidence level and a margin of error of $\pm 5\%$.
- a) Determine the minimum sample size required to conduct the study.
 - b) Apply the finite population correction to obtain the adjusted sample size.

Section “C”

Long Answer Questions

Attempt any Two questions (2×15=30)

17. The following data show the relationship between the number of training hours completed by software developers and their coding performance score in an IT company.

Training Hours	10	25	30	35	15	20
Performance Score	55	50	40	30	60	65

- Calculate the Karl Pearson correlation coefficient between training hours and performance score.
 - Find the coefficient of determination and interpret.
 - Predict the performance score when training hours are 28 hours.
 - Interpret the meaning of constant and regression coefficient.
18. A software company wants to test whether the mean bug-fixing time (in minutes) differs across three debugging tools (Tool A, Tool B, Tool C). The following times were recorded below;

Tool A	Tool B	Tool C
2	8	6
8	7	5
4	2	3
6	—	4

At $\alpha = 5\%$, use one-way ANOVA to test whether there is a significant difference in the mean bug-fixing time among the three tools.

19. A) An IT company wants to evaluate whether a new coding training program improves programmers' performance. Coding test scores of 8 programmers were recorded before and after the training.

Before Training	After Training
65	72
70	75
68	74
60	66
72	78
66	70
64	69
69	73

Test whether the training program has significantly improved the programmers' performance using an appropriate test. Use $\alpha = 1\%$. **(10)**

B) A company believes that the average employee job satisfaction score should be at least 70 (on a 0–100 scale). A random sample of 12 employees produced the following satisfaction scores:

72, 68, 75, 70, 66, 74, 71, 69, 73, 67, 70, 72

Construct a 95% confidence interval for the population mean job satisfaction score. **(5)**

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